3 **Matthew Lusty**
Program Engineer, GE Aerospace

WINGED DRONES AND FLIGHT

Written by: _____ ☐ Coach checked Sent on: _____ ☐ Reply came back

To: Matthew Lusty **Subject:** _____

Hello **Matthew**!

We are a FIRST LEGO League (FLL) team called "Look It's A Giraffe", and we're looking for some help with our innovation project. We thought maybe you could answer some of our questions.

For our innovation project, our topic is invasive grass in rainforests. Specifically, how the grass is a threat to the rainforests.

Our questions for Matthew (one or two great questions beat five okay ones; ask about what he knows)

1. what is the best drone, winged or helicopter? wich is better for weight and distance? (w/d ratio)
2. could winged droner carry water? How much?
3. How could we improve our idea?

Thanks
Look It's A Giraffe
Sylvie, Jordan, Jack, Ella, Brynn, Fedor, Ira, Alexis, Amelia
Team Coach: Chris Leigh

About Matthew

- Program Engineer at GE Aerospace, a company that makes engines and parts for aircraft.
- An **aerospace engineer**, so he knows how things fly. He can tell us about **winged (fixed-wing) drones**.
- Suggested by Sylvie.

Ideas to ask about: winged vs copter drones: range and flight time · carrying and dropping a payload · take-off and landing in a rainforest · flying in heat, smoke and wind

2 Dan Panko

Manager, Ontario Ministry of the Environment, Conservation and Parks

ENVIRONMENT AND PESTICIDE RULES

Written by: _____ ☐ Coach checked Sent on: _____ ☐ Reply came back

To: Dan Panko **Subject:** Logo League

Hello **Dan**!

We are a FIRST LEGO League (FLL) team called "Look It's A Giraffe", and we're looking for some help with our innovation project. We thought maybe you could answer some of our questions.

For our innovation project, our topic is invasive grass in rainforests. Specifically, how the grass is a threat to the rainforests.

Our questions for Dan (one or two great questions beat five okay ones; ask about what he knows)

1. how can we prevent the grass from growing in rainforests area

2. What Do you think of our solution. ~~for preventing~~

3. ~~Do you invasive grass plan~~ have you ever Delt with invasive grass, how do you Deal with it.

Thanks
Look It's A Giraffe
Sylvie, Jordan, Jack, Ella, Brynn, Fedor, Ira, Alexis, Amelia
Team Coach: Chris Leigh

About Dan

- Manager at the Ontario Ministry of the Environment, Conservation and Parks, with over 22 years in government.
- Leads technical and science work on **air quality**, **environmental assessment**, **pesticide rules** and making sure people follow environmental laws. He is a Designated Provincial Officer.
- Expert in **environmental sampling** (testing air, soil or water to find a problem) and in designing government programs.
- Knows how to teach, so he is good at explaining things.

Ideas to ask about: herbicide safety near native plants · rules for drones that spray · sampling to track where grass spreads · smoke and air quality from fires · invasive plants here in Ontario

1 **Bob Leigh, P.Eng.**
Senior Director of Commercial Markets, RTI

DRONES THAT THINK FOR THEMSELVES

Written by: _____ ☐ Coach checked Sent on: _____ ☐ Reply came back

To: Bob Leigh **Subject:** _____

Hello **Bob**!

We are a FIRST LEGO League (FLL) team called "Look It's A Giraffe", and we're looking for some help with our innovation project. We thought maybe you could answer some of our questions.

For our innovation project, our topic is invasive grass in rainforests. Specifically, how the grass is a threat to the rainforests.

Our questions for Bob (one or two great questions beat five okay ones; ask about what he knows)

1. what do you think of our solution to invasive grass causing fires in rainforests? Our solution is to get an image from satelites of where the invasive grass in the rainforest is located.
2. What is an autonomous system and do you think it could benefit our solution?
3. What sensors do you think our drone could use/need?

Thanks
Look It's A Giraffe
Sylvie, Jordan, Jack, Ella, Brynn, Fedor, Ira, Alexis, Amelia
Team Coach: Chris Leigh

About Bob

- Professional Engineer (P.Eng.). Senior Director of Commercial Markets at RTI.
- Knows about **autonomous systems**: machines like drones that can find their way and make decisions without a person steering them.
- Works on problems nobody has solved yet.
- Suggested by Jordan.

Ideas to ask about: drones that fly and map on their own · telling invasive grass apart from native plants · predicting which spots will burn next · what makes autonomous drones hard to build

4 **Expert:** PETE

Job and where they work: firefighter

NEW EXPERT

Written by: Feodh and IRA ☐ Coach checked Sent on: ☐ Reply came back

To: Pete **Subject:** Questions

Hello Pete !

We are a FIRST LEGO League (FLL) team called "Look It's A Giraffe", and we're looking for some help with our innovation project. We thought maybe you could answer some of our questions.

For our innovation project, our topic is invasive grass in rainforests. Specifically, how the grass is a threat to the rainforests.

Our questions (one or two great questions beat five okay ones; ask about what they know)

1. Do you know if firefighters use copter or winged drones in their work?
2. What makes firefighting difficult in rainforests?
3. Our solution is winged drones that drop water on fire is that a good solution?

Thanks
Look It's A Giraffe
Sylvie, Jordan, Jack, Ella, Brynn, Fedor, Ira, Alexis, Amelia
Team Coach: Chris Leigh

About this expert

What they do and what they know about:

Why we picked them, and who suggested them: